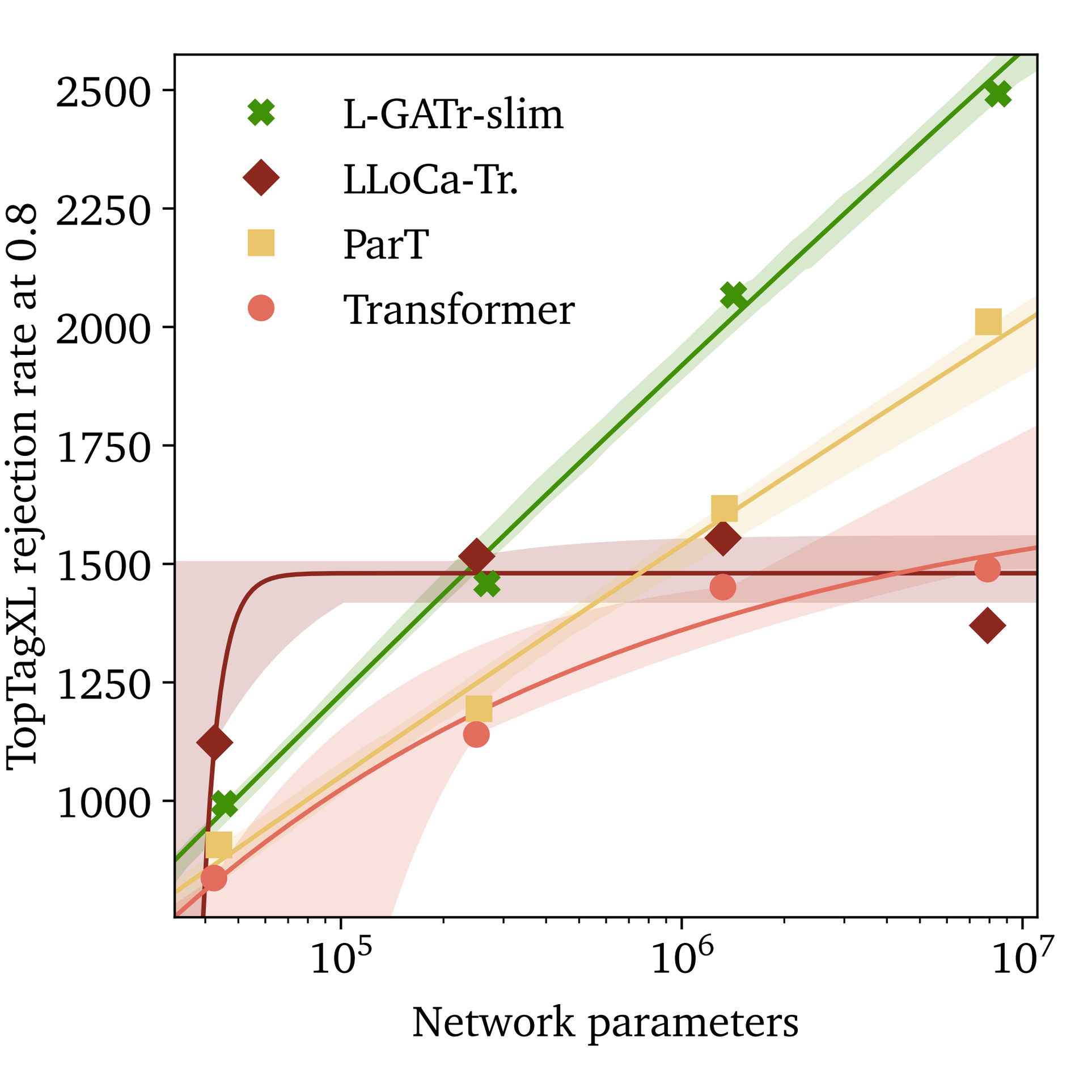
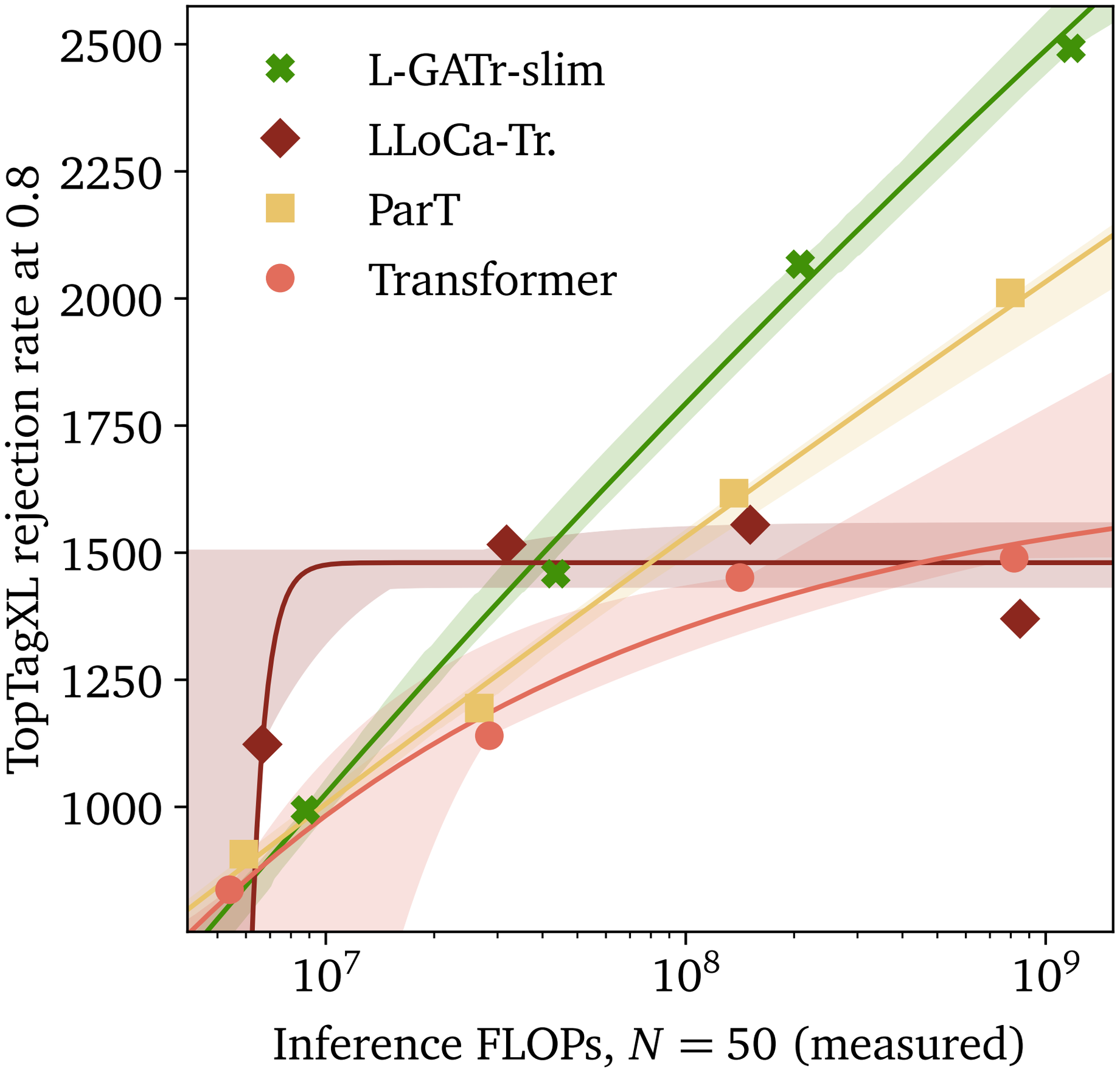
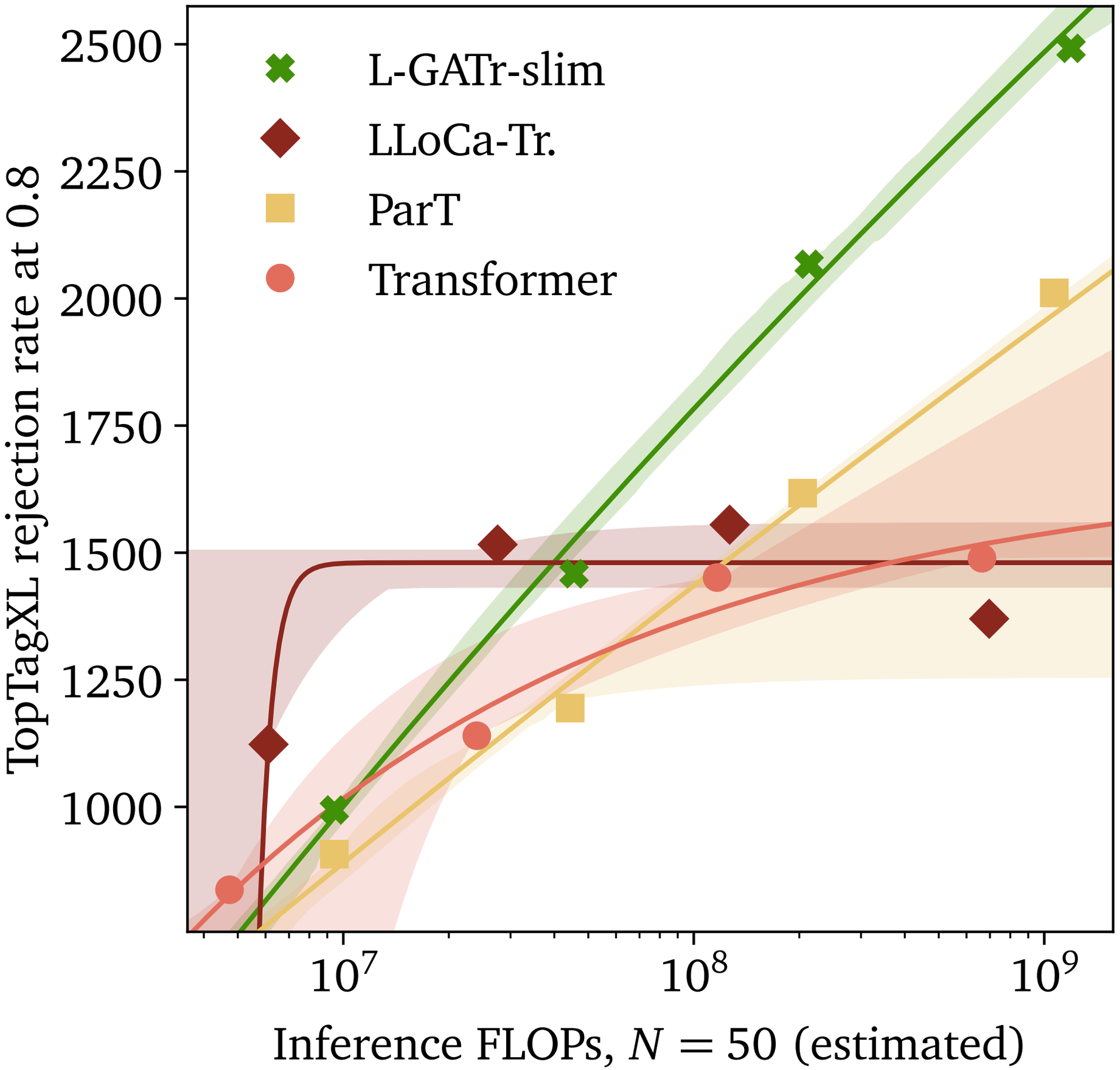
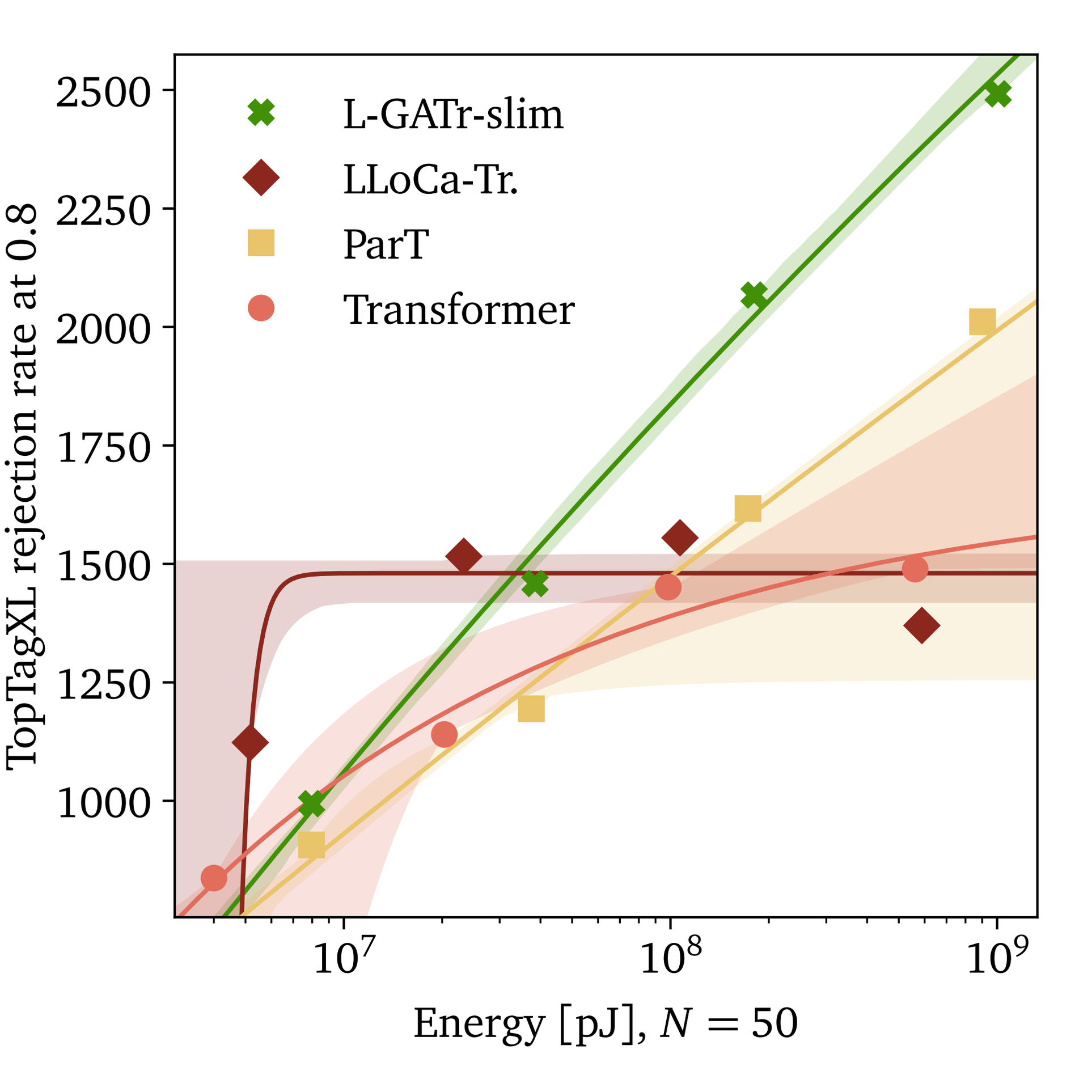
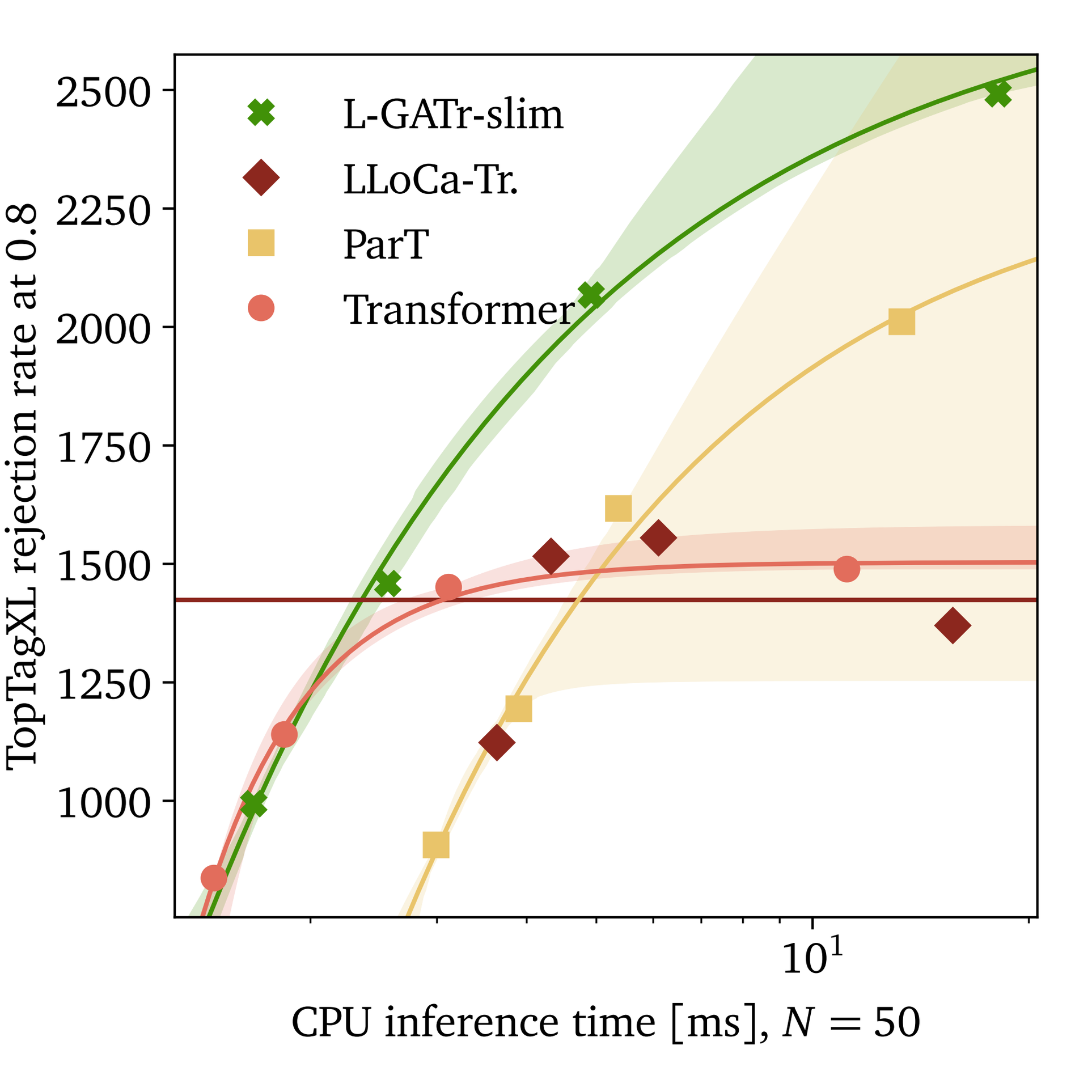
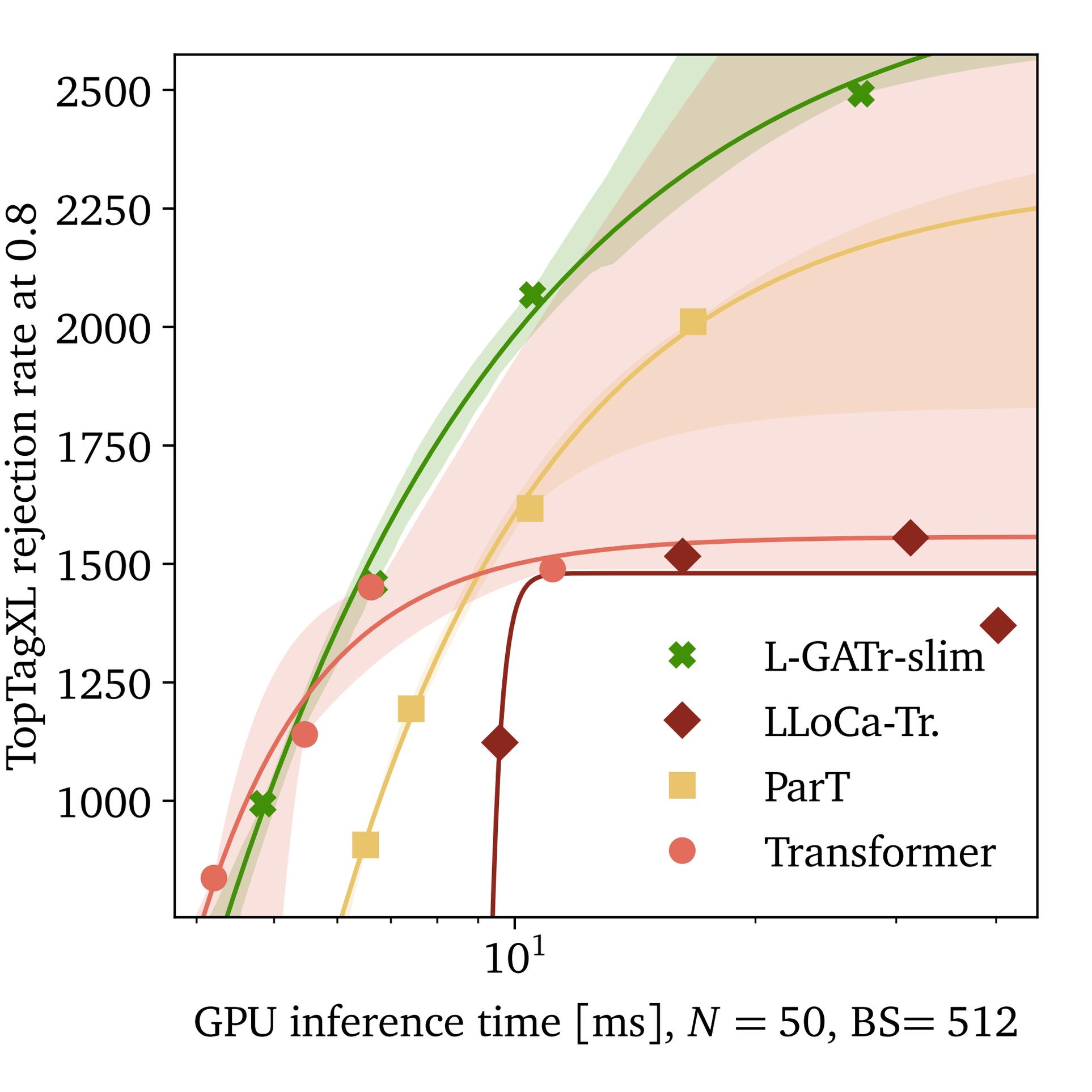

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
TopTagXL rejection rate at 0.8
2500
2250
2000
1750
1500
1250
1000
10^5
10^6
10^7
Network parameters

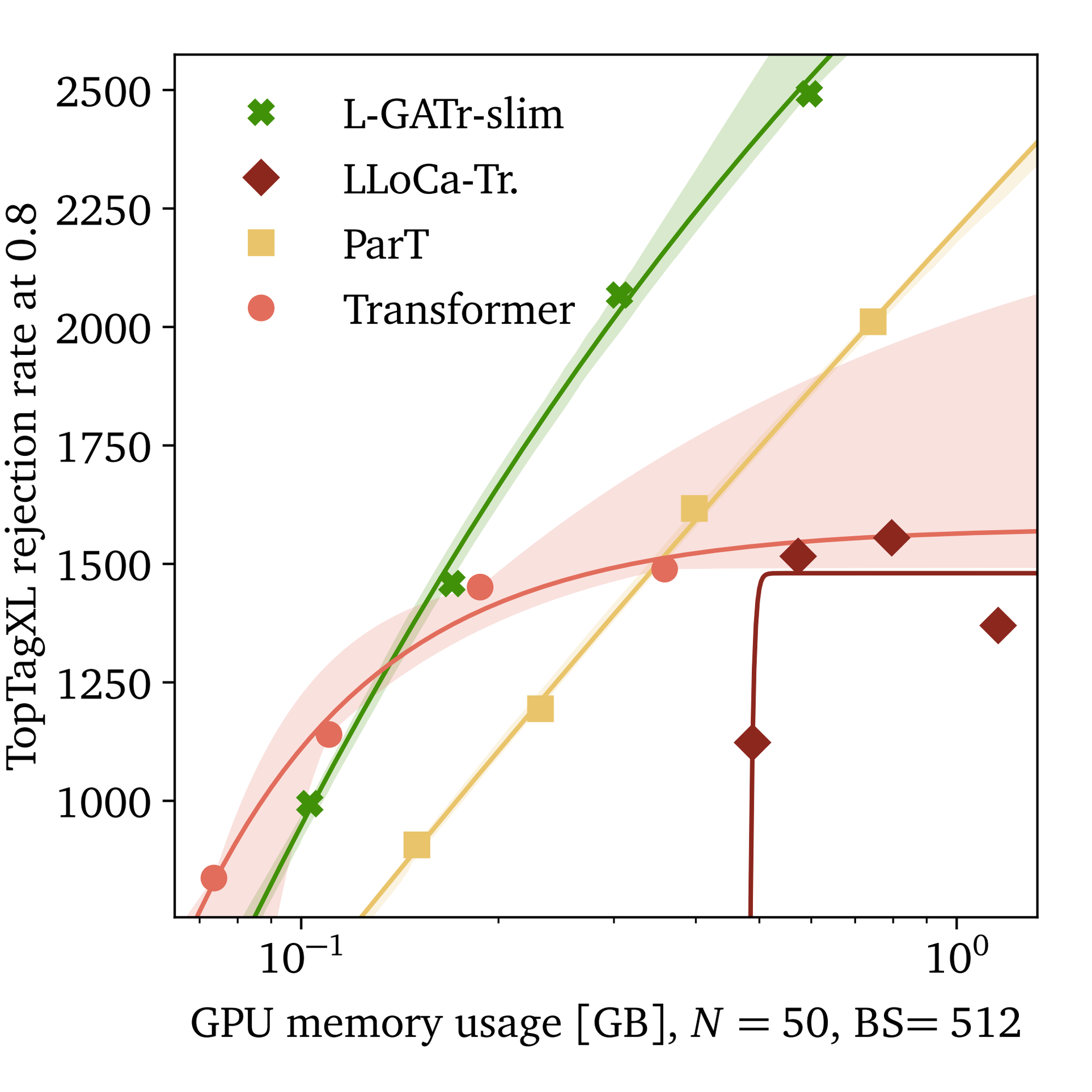

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
TopTagXL rejection rate at 0.8
2500
2250
2000
1750
1500
1250
1000
10^7
10^8
10^9
Inference FLOPs, $N = 50$ (measured)

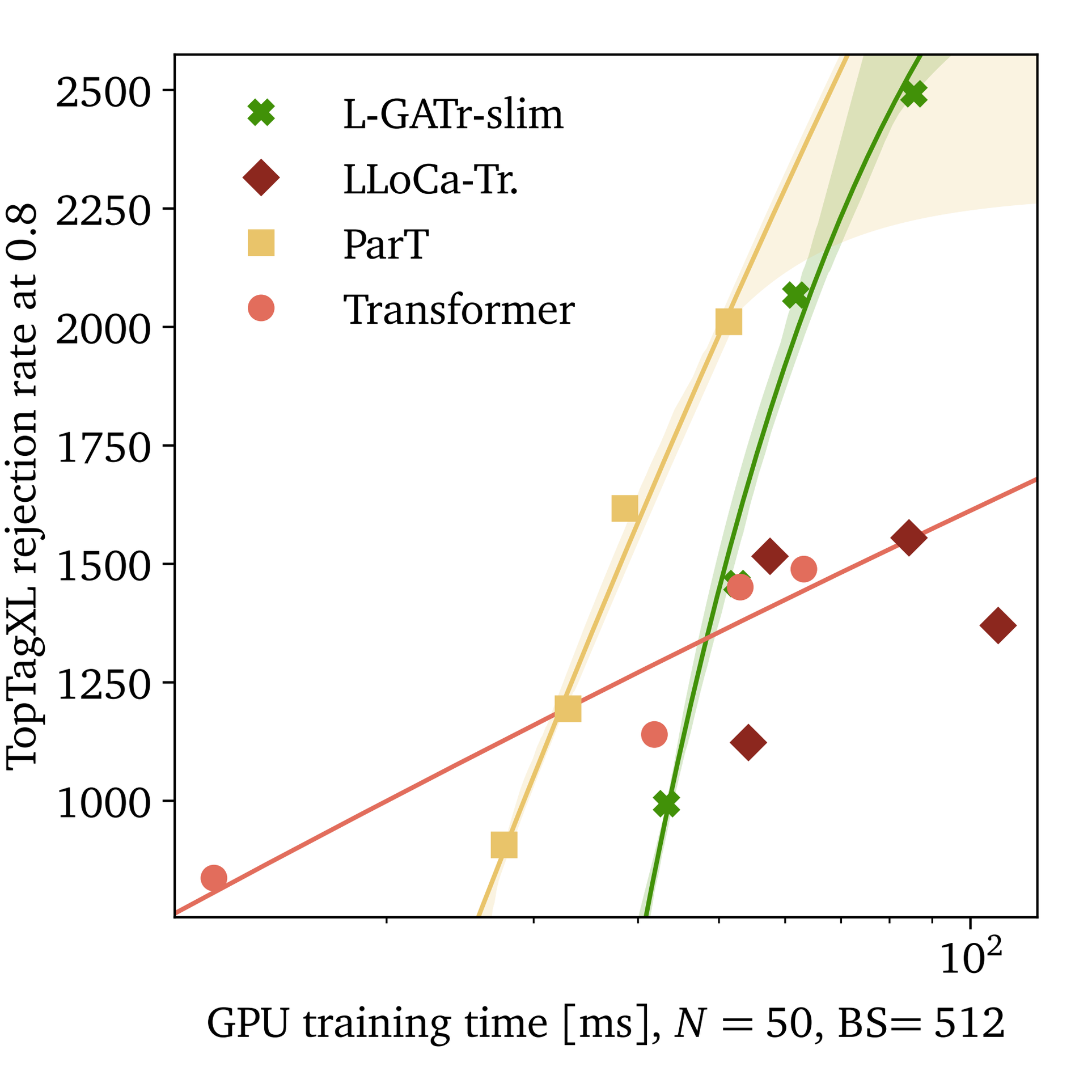

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
TopTagXL rejection rate at 0.8
2500
2250
2000
1750
1500
1250
1000
10^7
10^8
10^9
Inference FLOPs, $N = 50$ (estimated)

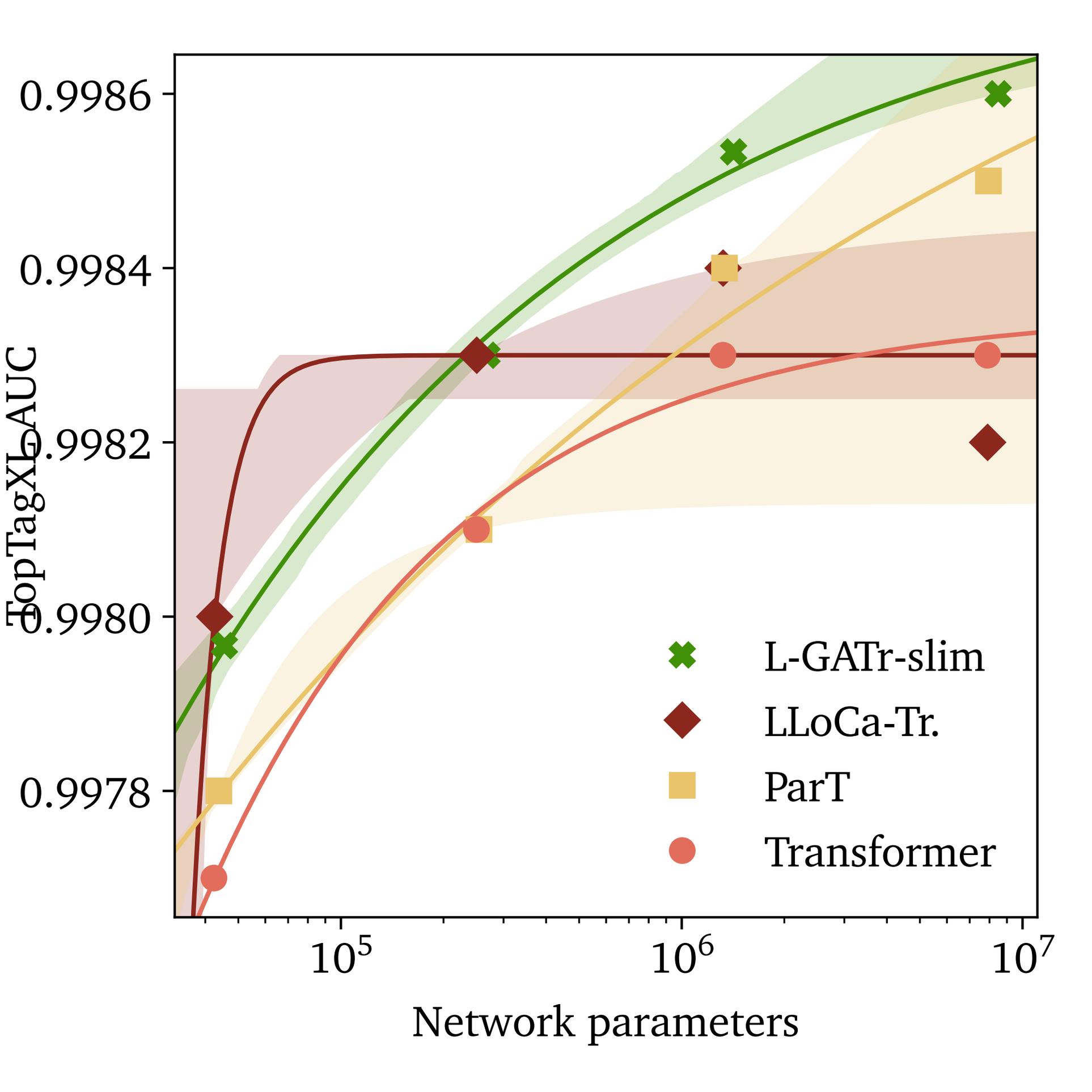

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
TopTagXL rejection rate at 0.8
2500
2250
2000
1750
1500
1250
1000
10^7
10^8
10^9
Energy [pJ], $N = 50$

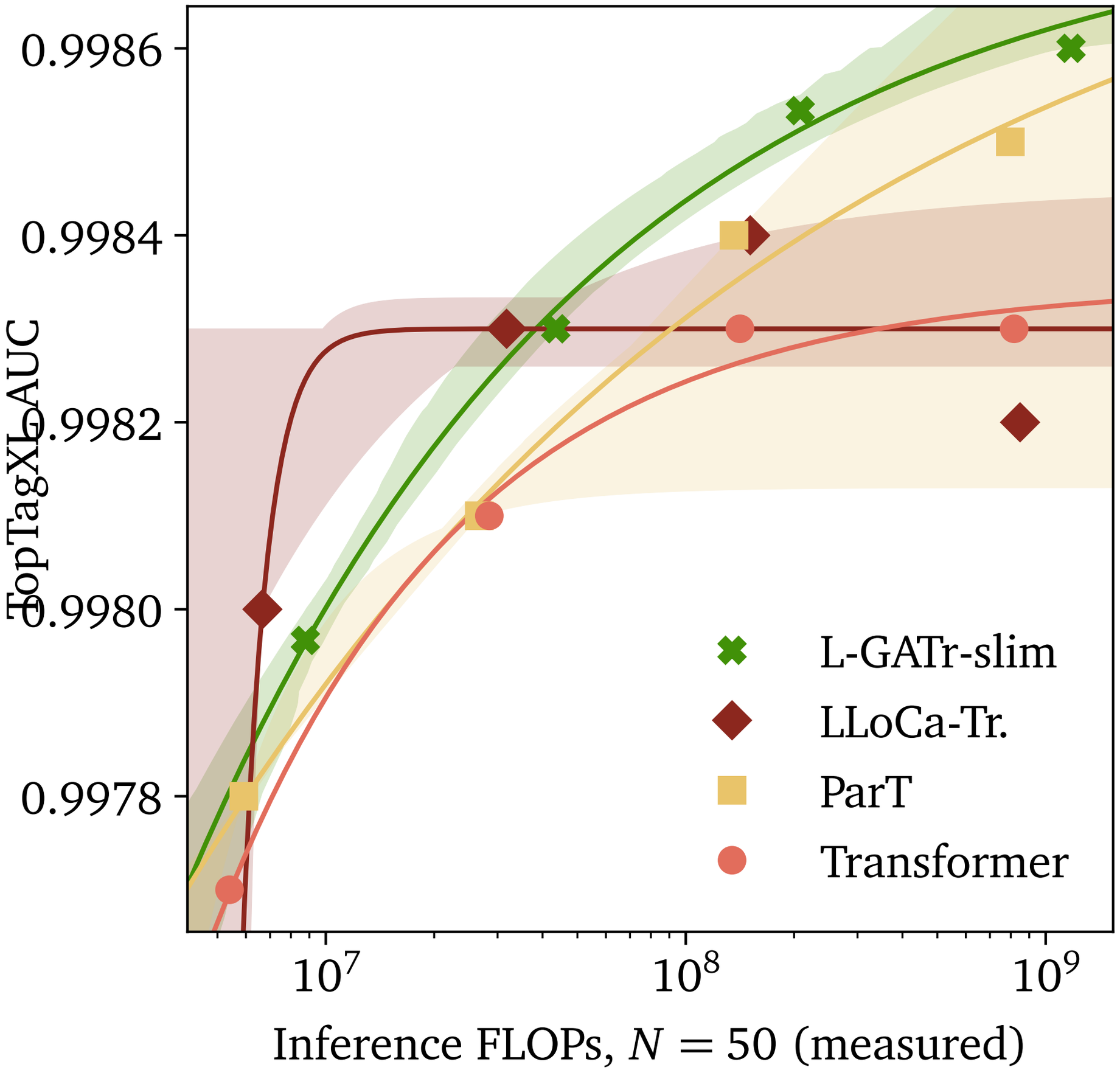

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2500
2250
2000
1750
1500
1250
1000
TopTagXL rejection rate at 0.8
10^1
CPU inference time [ms], $N = 50$

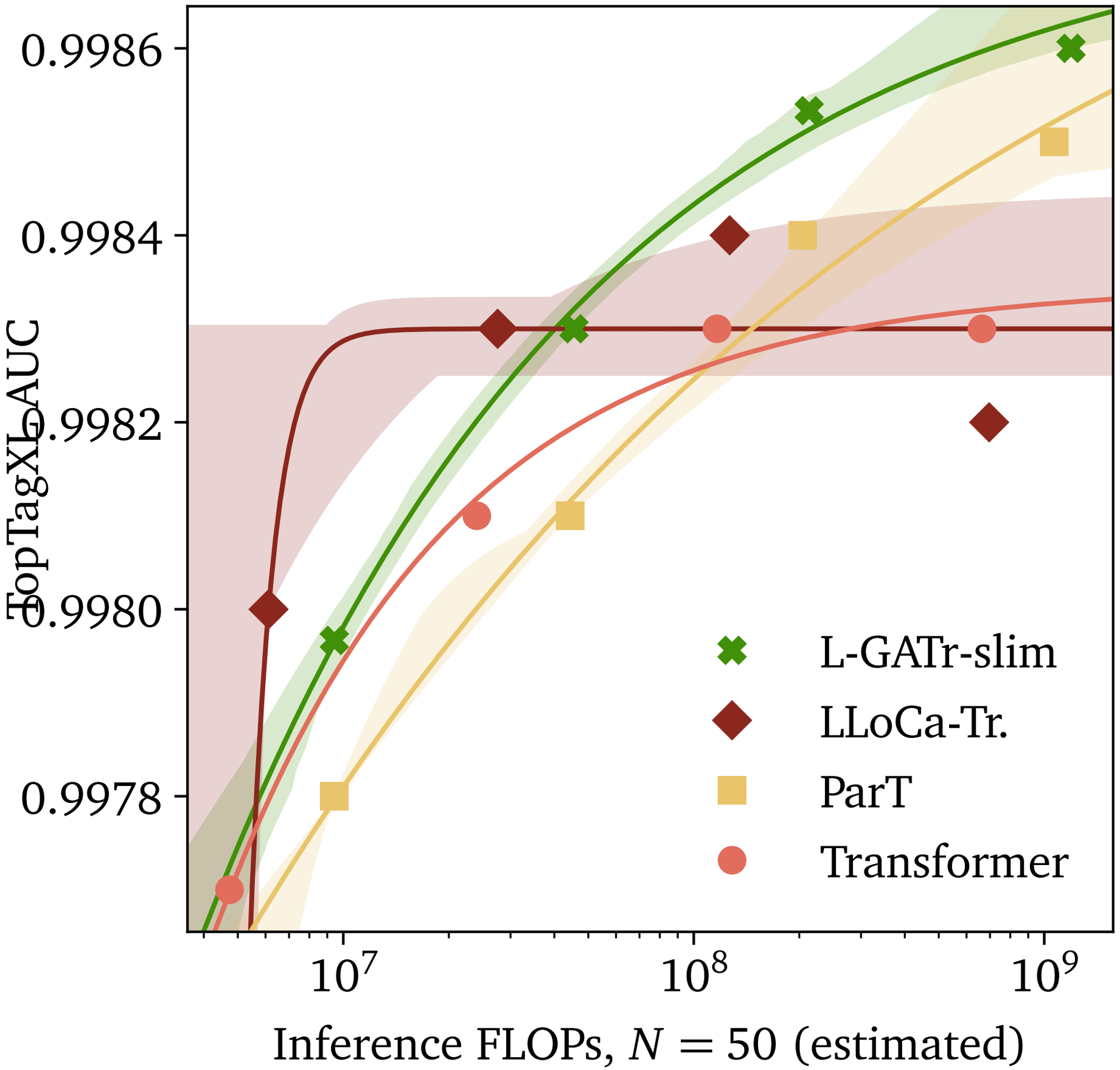

2500
2250
2000
1750
1500
1250
1000
TopTagXL rejection rate at 0.8
10^1
GPU inference time [ms], $N = 50$, BS$= 512$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

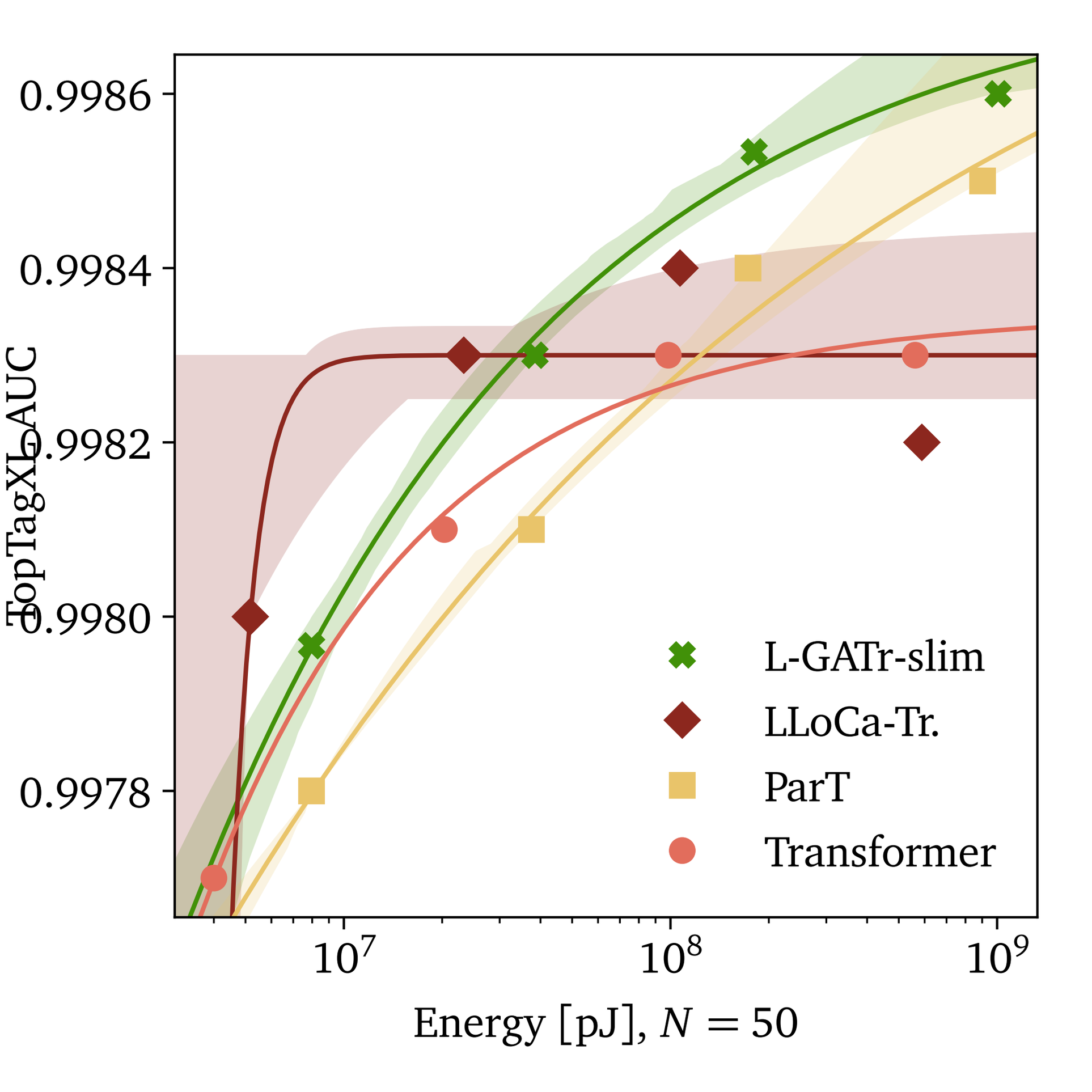

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2500
2250
2000
1750
1500
1250
1000
TopTagXL rejection rate at 0.8
10^{-1}
10^{0}
GPU memory usage [GB], $N = 50$, BS$= 512$

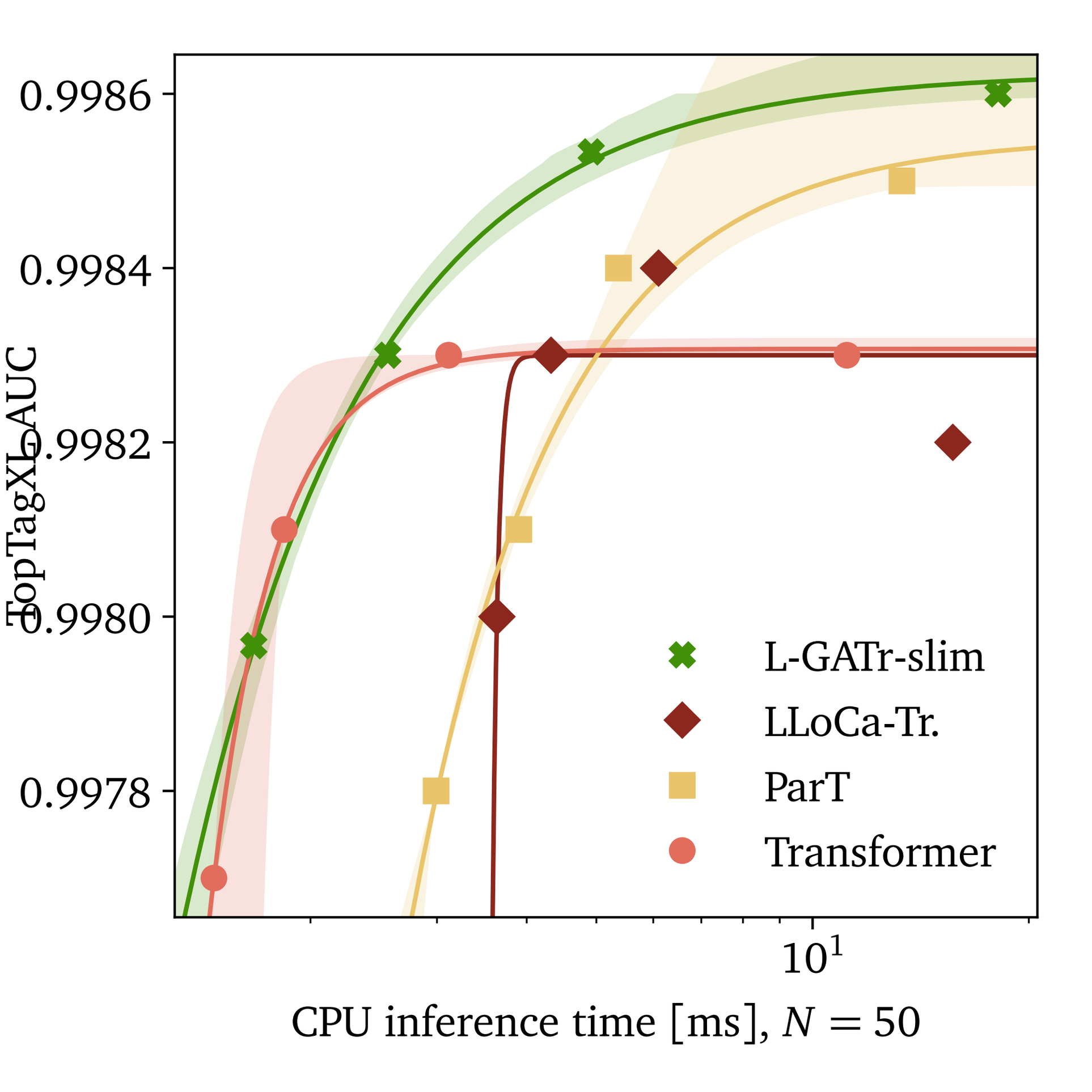

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2500
2250
2000
1750
1500
1250
1000
TopTagXL rejection rate at 0.8
10^2
GPU training time [ms], $N = 50$, BS$= 512$

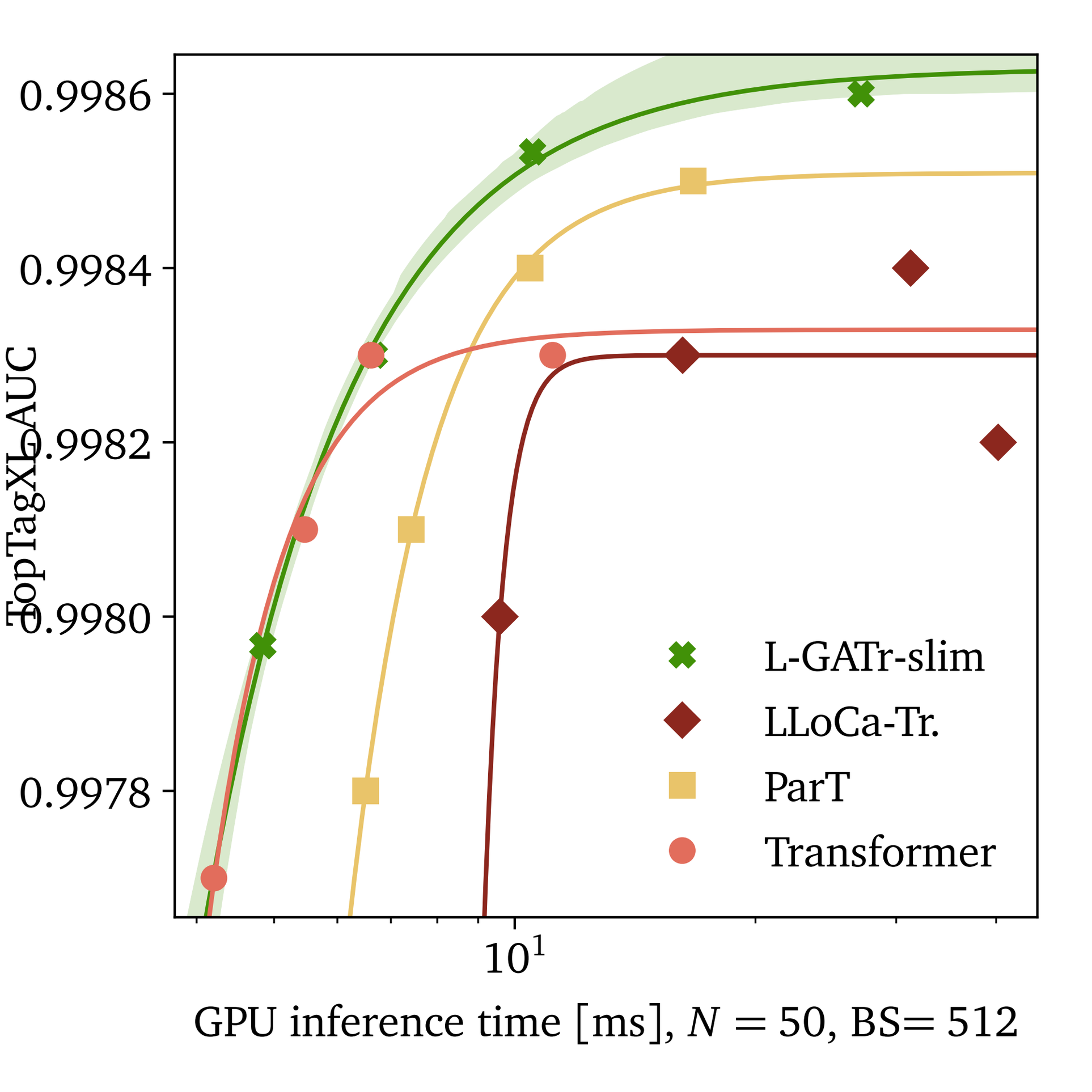

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXLoAUC
10^5
10^6
10^7
Network parameters
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

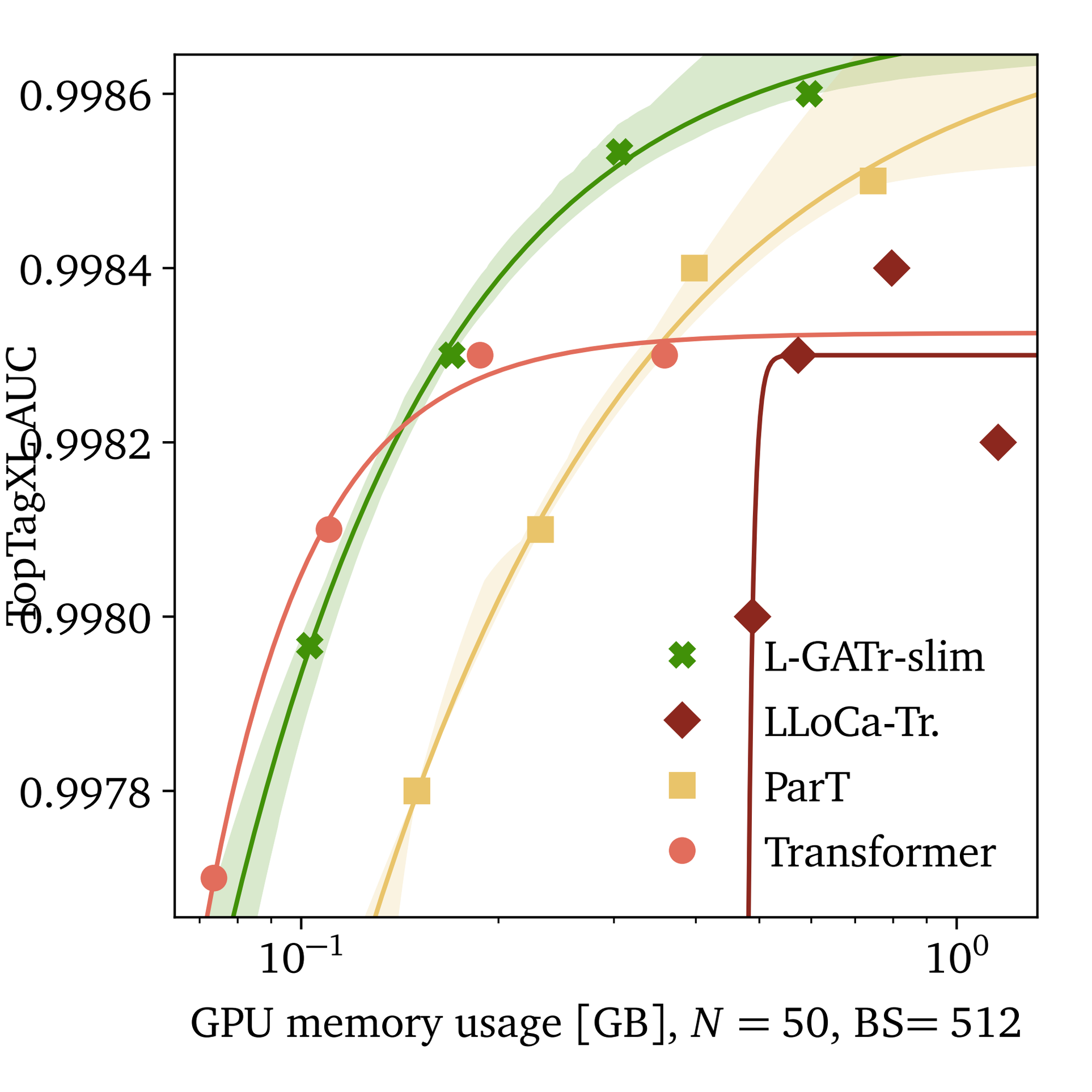

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXbb AUC
10^7
10^8
10^9
Inference FLOPs, $N = 50$ (measured)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

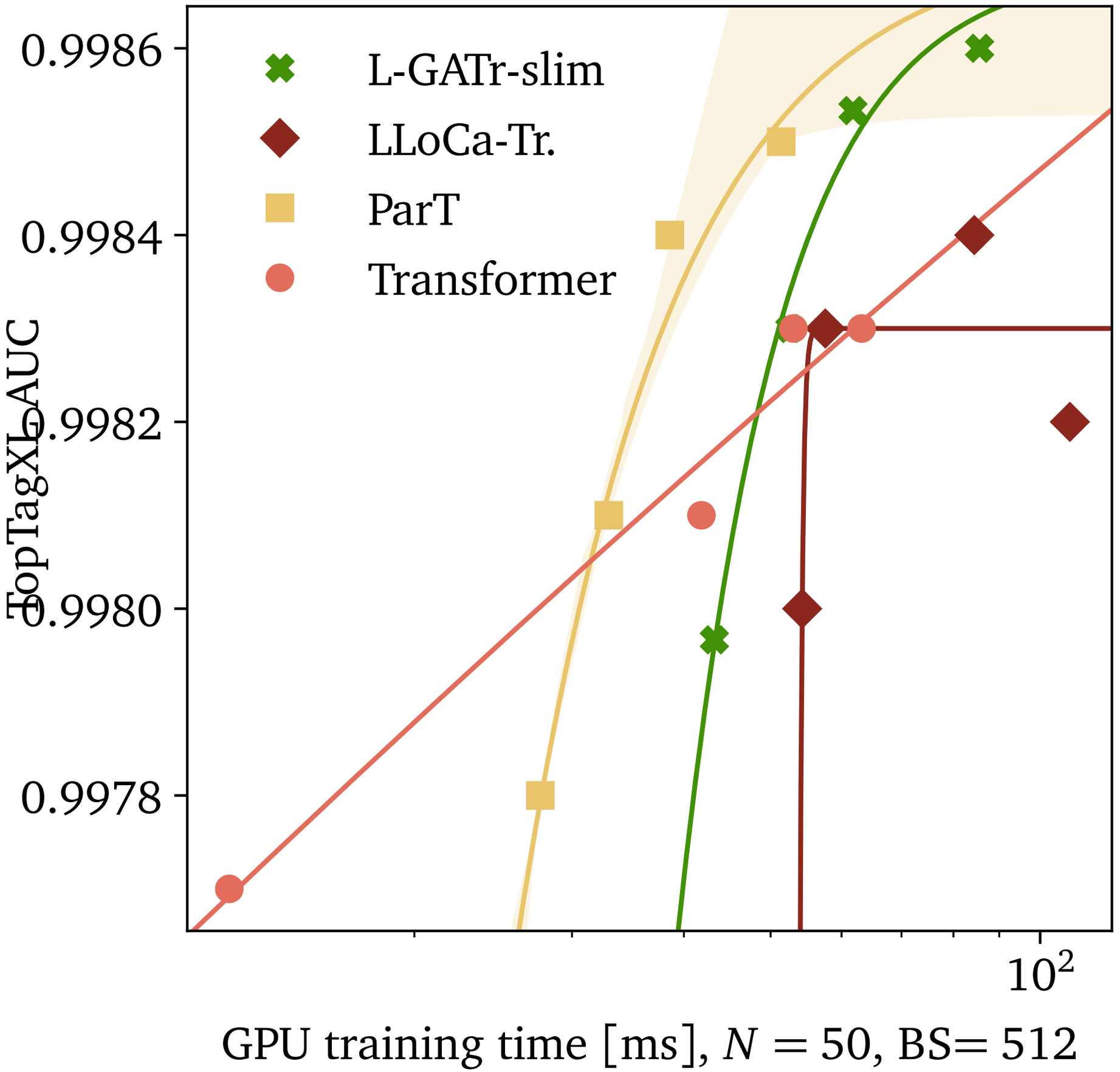

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXbb AUC
10^7
10^8
10^9
Inference FLOPs, $N = 50$ (estimated)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

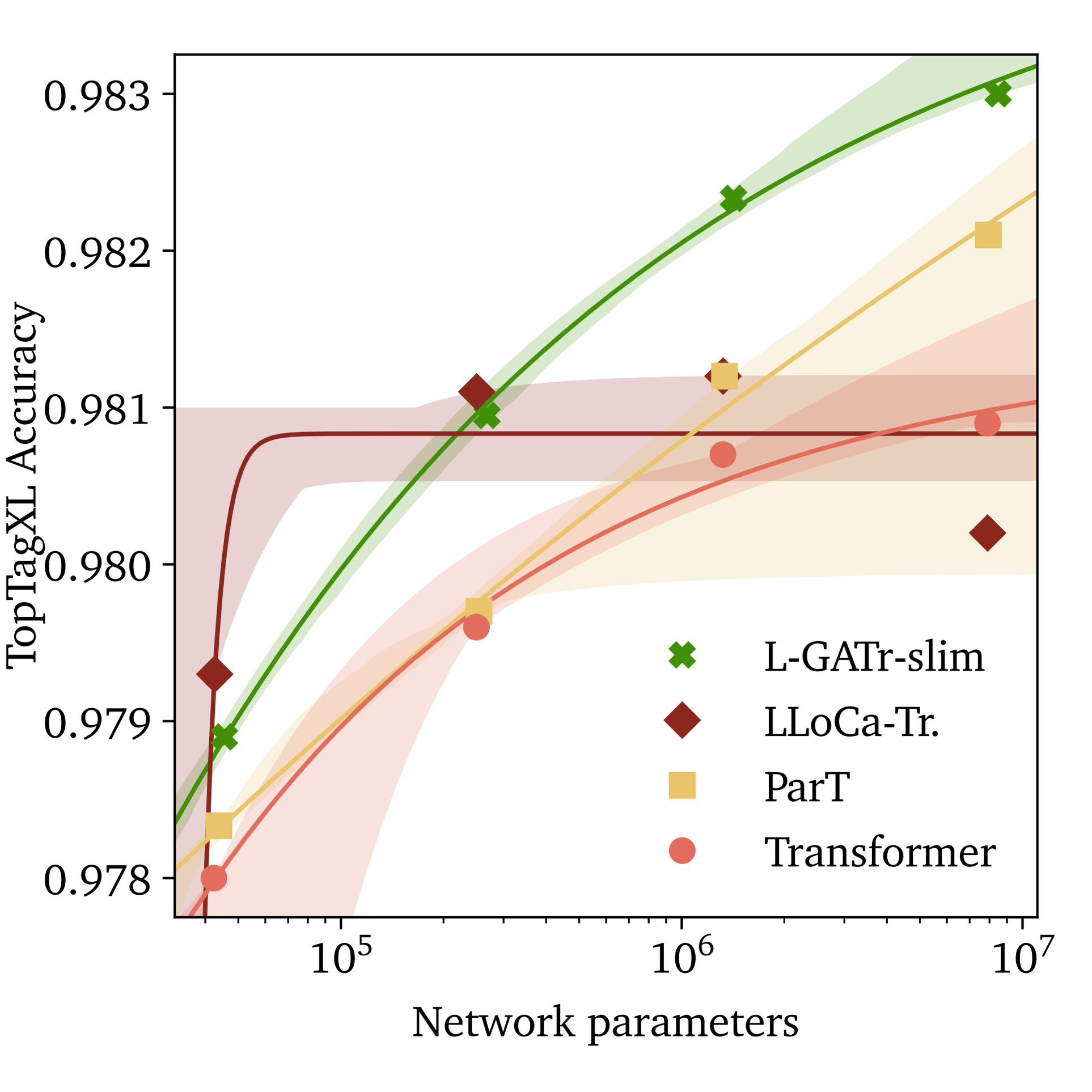

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXbb AUC
10^7
10^8
10^9
Energy [pJ], $N = 50$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

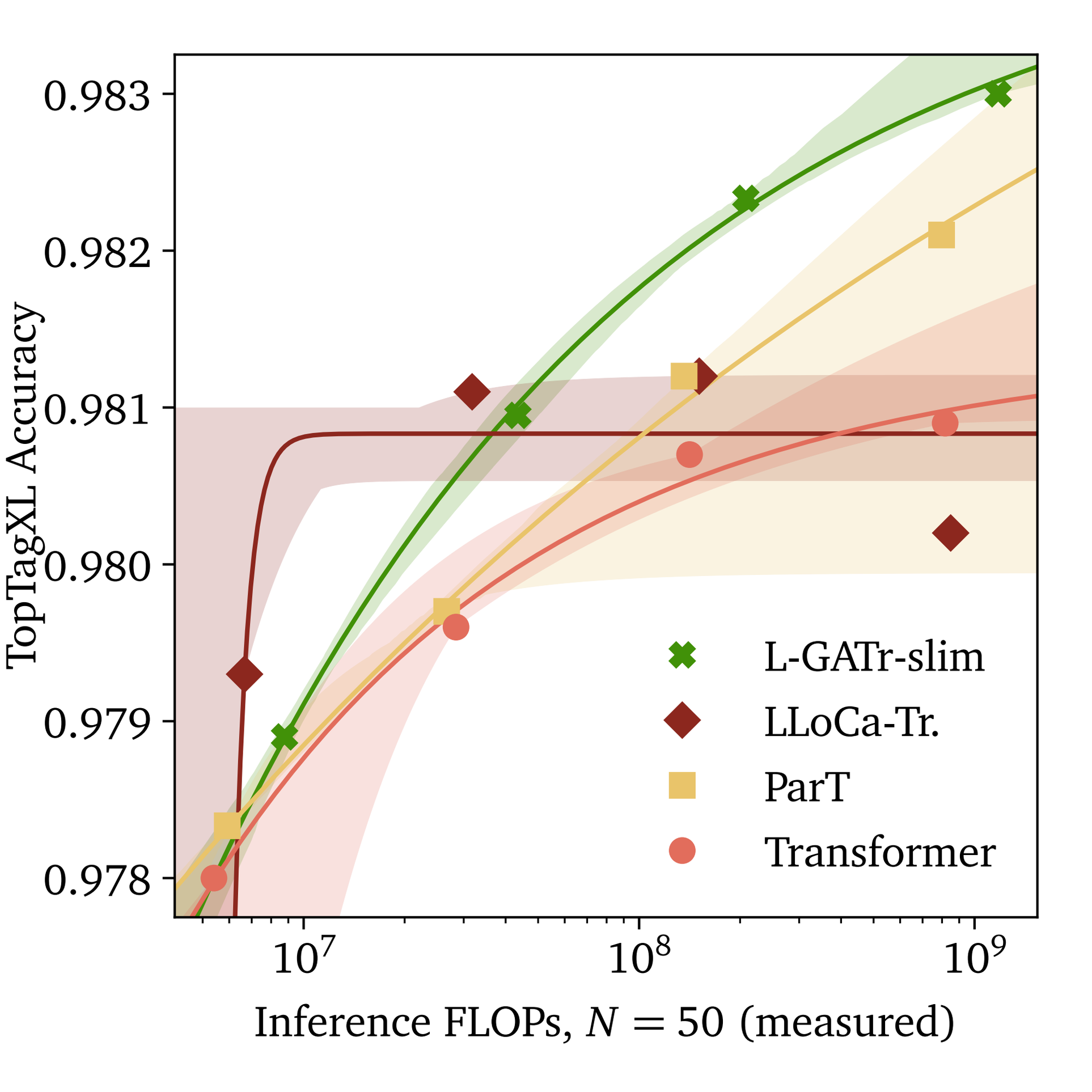

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXbb AUC
10^1
CPU inference time [ms], $N = 50$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

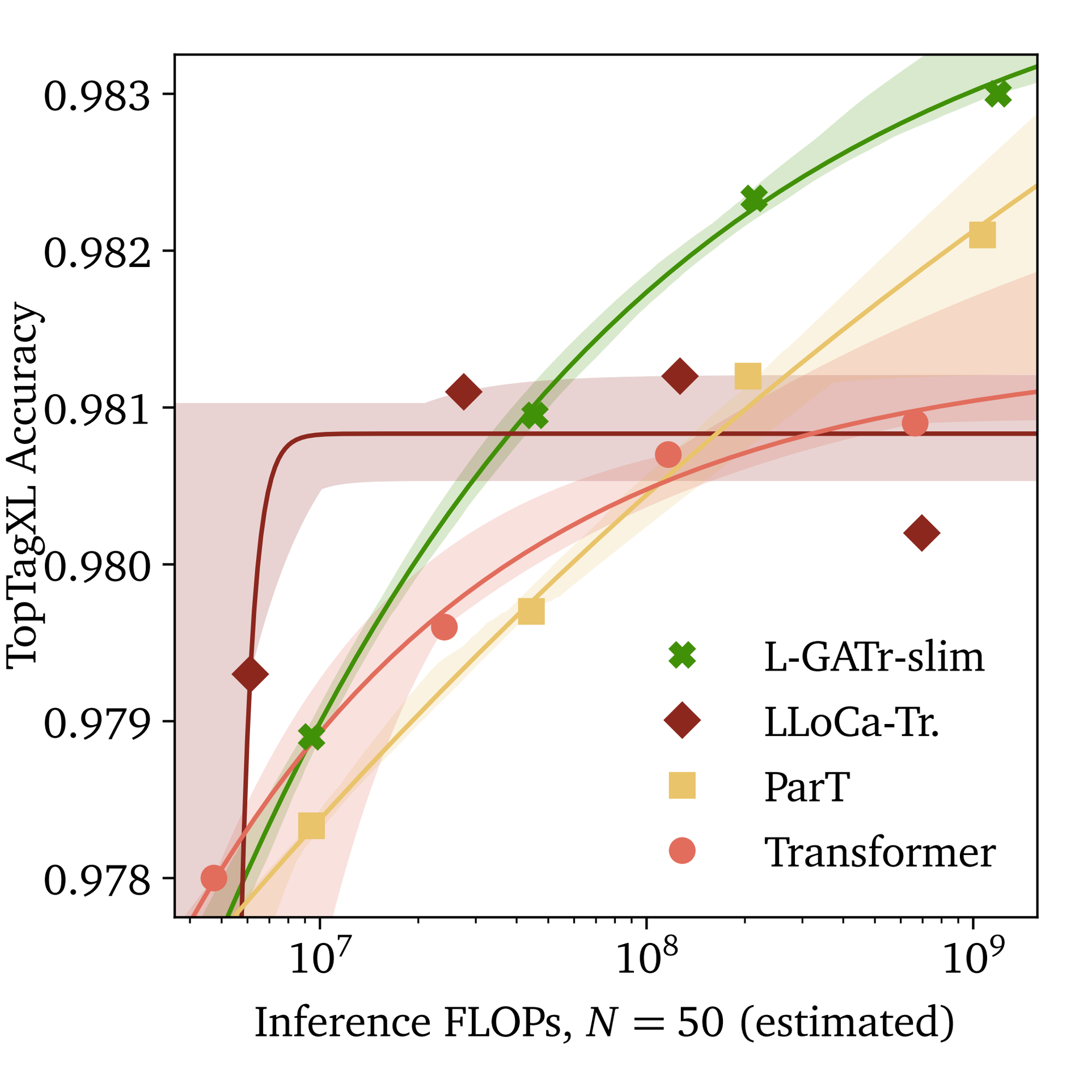

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXbAUC
10^1
GPU inference time [ms], $N = 50$, BS= 512
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

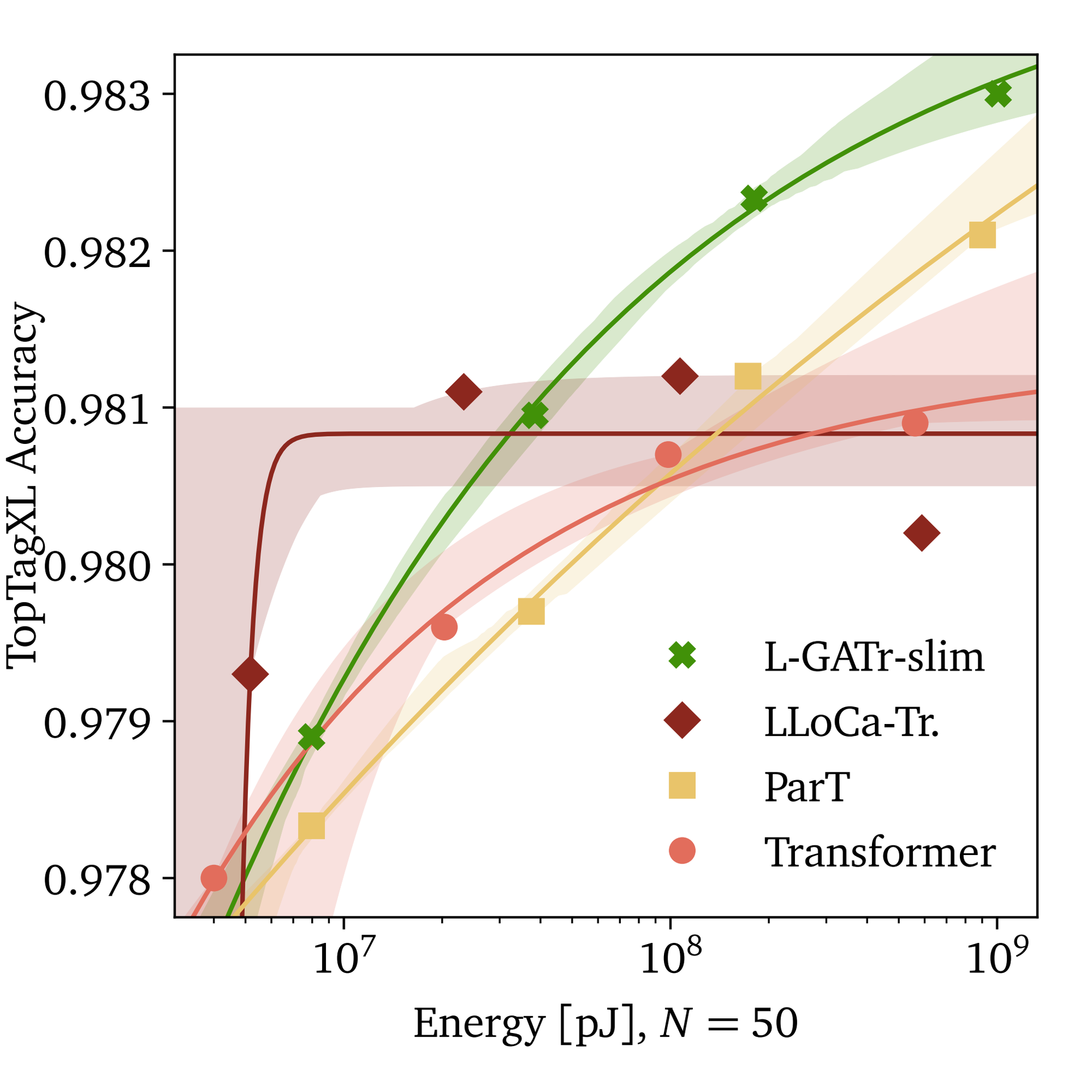

0.9986
0.9984
0.9982
0.9980
0.9978
TopTagXbb AUC
10^{-1}
10^{0}
GPU memory usage [GB], $N = 50$, BS$= 512$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

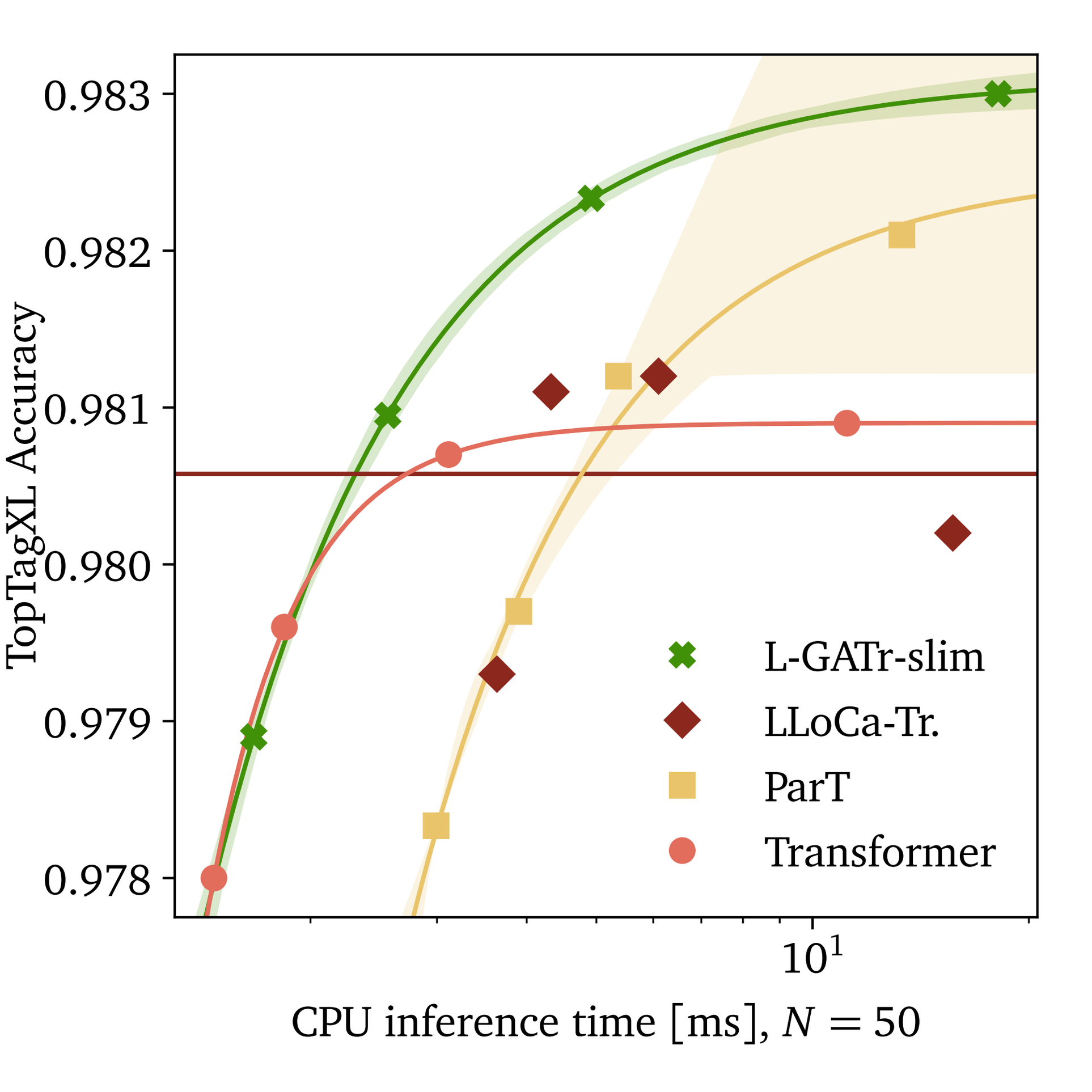

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
0.9986
0.9984
0.9982
0.9980
0.9978
10^2
GPU training time [ms], $N = 50$, BS$= 512$

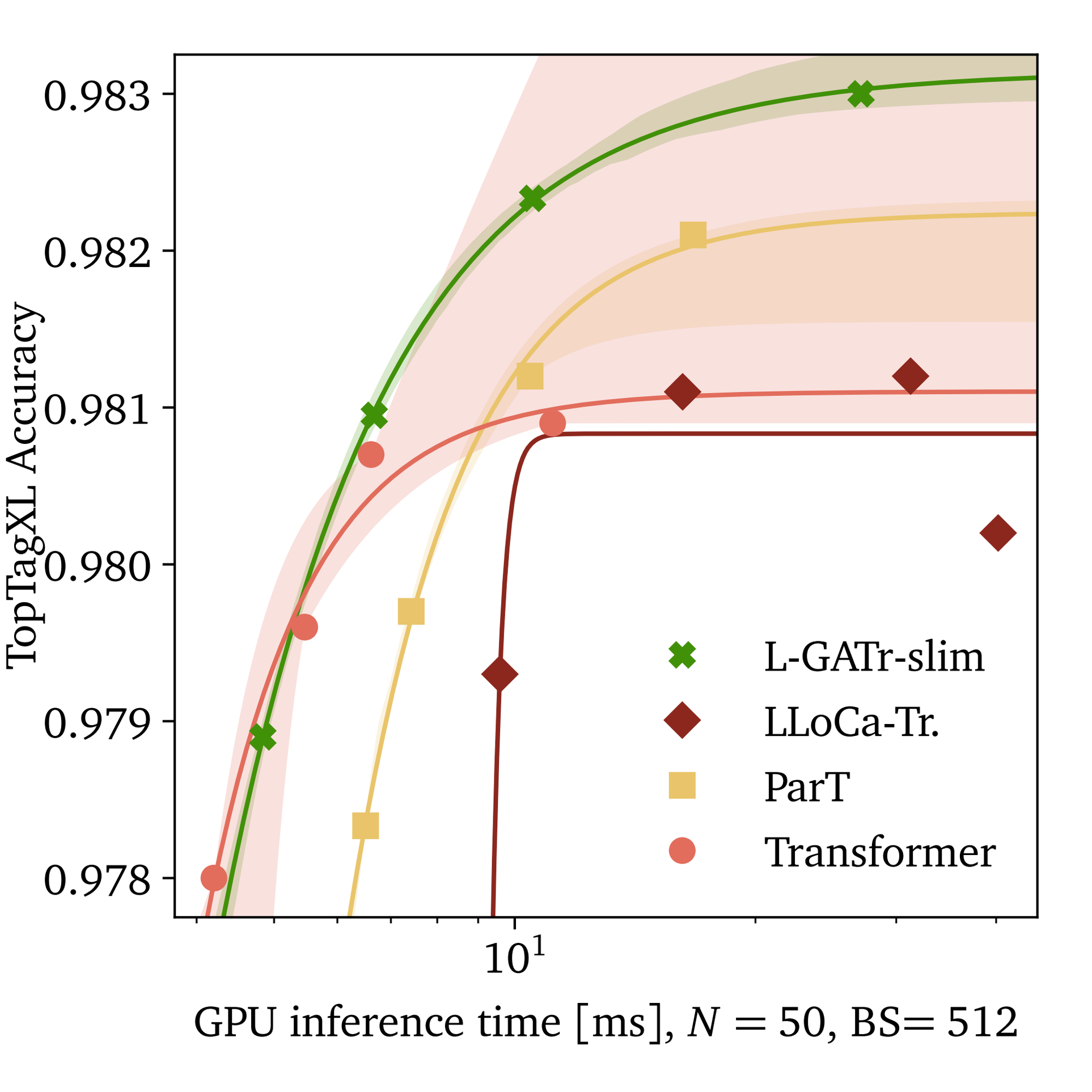

0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^5
10^6
10^7
Network parameters
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

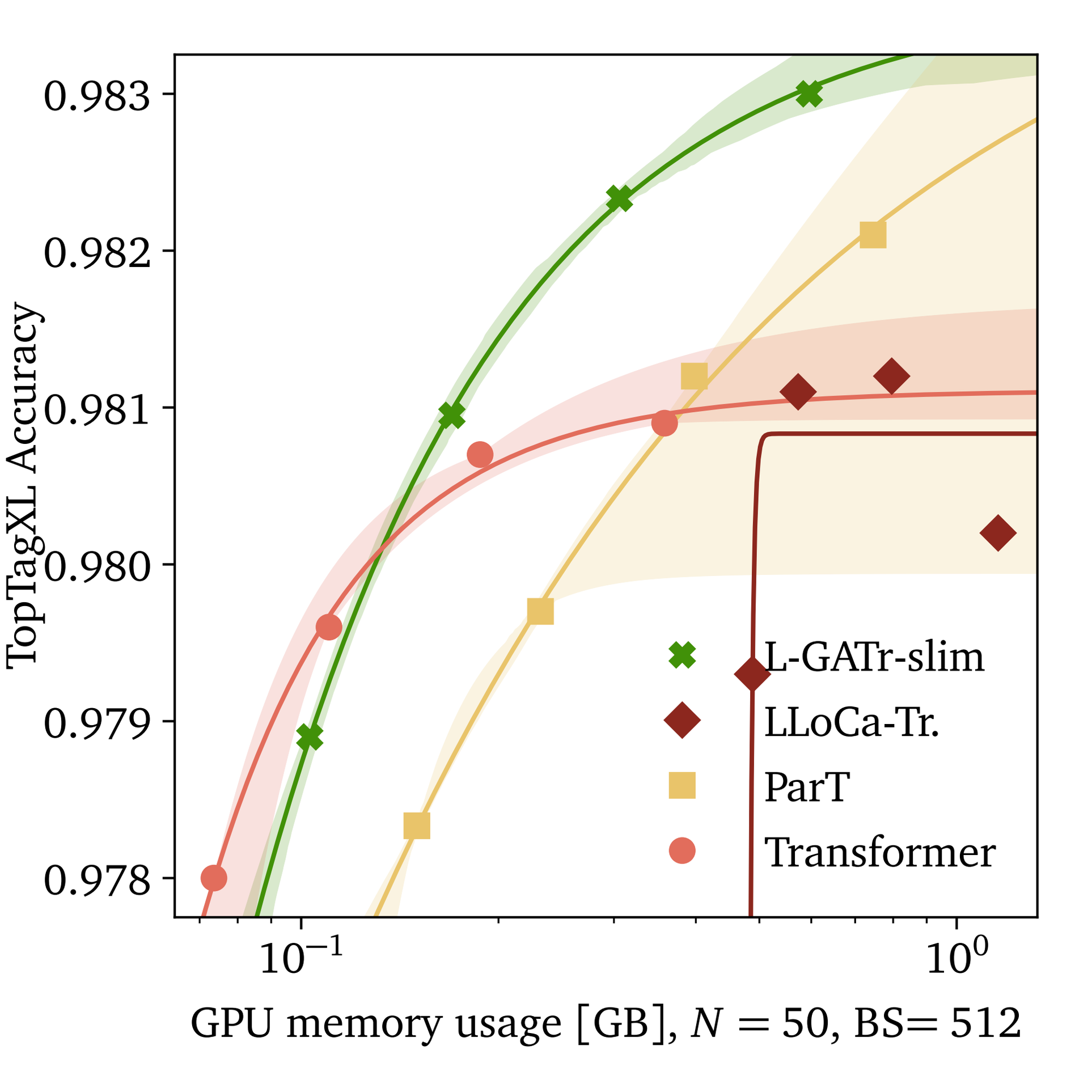

0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^7
10^8
10^9
Inference FLOPs, $N = 50$ (measured)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

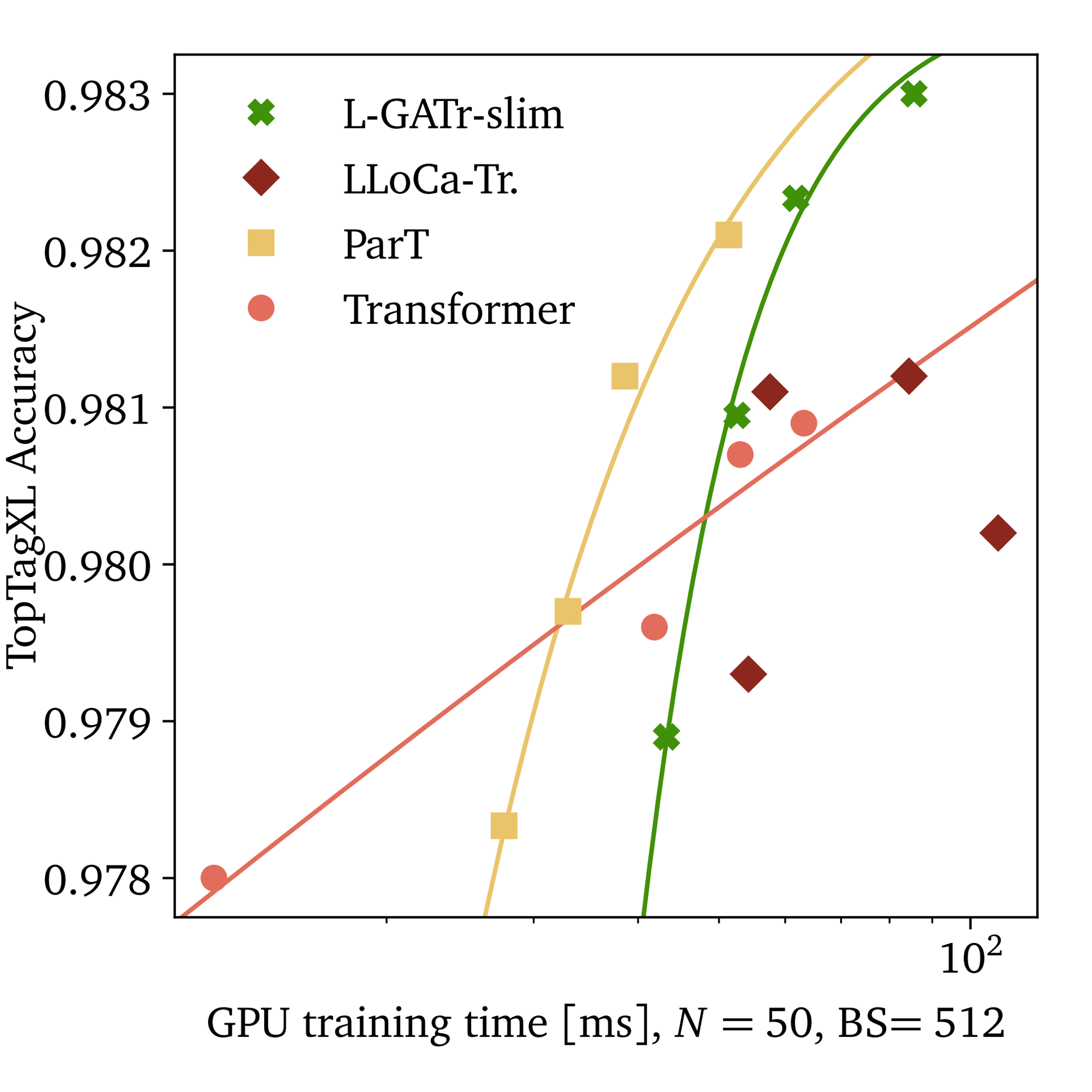

0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^7
10^8
10^9
Inference FLOPs, $N = 50$ (estimated)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^7
10^8
10^9
Energy [pJ], $N = 50$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^1
CPU inference time [ms], $N = 50$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^1
GPU inference time [ms], $N = 50$, BS$= 512$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^{-1}
10^{0}
GPU memory usage [GB], $N = 50$, BS$= 512$
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


L-GATr-slim
LLoCa-Tr.
ParT
Transformer
0.983
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^2
GPU training time [ms], $N = 50$, BS$= 512$